Fiscal Year:	FY 2020	**Task Last Updated:**	FY 02/17/2020
PI Name:	Hyde, Truell Ph.D.		
Project Title:	PK-4: Self-Ordering of Interacting Complex Plasma Particles in Microgravity		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	FUNDAMENTAL PHYSICS--Fundamental physics		
Joint Agency Name:	National Science Foundation	**TechPort:**	No
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Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	76798-7283	**Congressional District:**	17
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2016 Physical Sciences NNH16ZTT002N: Fundamental Physics – NASA/NSF Partnership on Science of Dusty Plasmas: Utilizing the PK-4 Facility on board the ISS
Start Date:	04/19/2017	**End Date:**	04/19/2021
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No. of PhD Candidates:	8	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	4	**Monitoring Center:**	NASA JPL
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Flight Program:	ISS		
Flight Assignment:	ISS		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Matthews, Lorin Ph.D. (Baylor University) Hartmann, Peter Ph.D. (Baylor University) Petrov, Oleg Ph.D. (Joint Institute for High Temperatures, Russian Academy of Sciences) Rosenberg, Marlene Ph.D. (University Of California, San Diego)		
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Task Description:	Complex plasma consists of ions, electrons, neutral gas, and solid microparticles referred to as dust. Dust grains immersed in plasma become negatively charged and interact with one another as well as plasma components through electromagnetic interactions. In particular, dust grains experience a plasma drag force, altering the trajectories of ions in the plasma, and participate in collective plasma interactions, such as waves and instabilities. An interesting aspect of complex plasma is its ability to self-organize into dust liquids, 2D and 3D dust crystal lattices, and 1D dust chains, which are aligned with electric fields in the plasma. This feature provides researchers the ability to examine classical collective phenomena on size and time scales allowing for direct observation (at the kinetic level) of fundamental microscopic and macroscopic processes, as the dust can easily be imaged with high-speed cameras. In many environments, positive ions in the plasma have a directed flow with respect to the negatively charged dust grains. The resulting interaction between the dust and flowing plasma creates an ion wakefield downstream from the dust particles, with the positive space region modifying the interaction between the charged grains and contributing to the observed dynamics and particle structure. Since the ion wakefield is weak, the effects are usually masked by gravity for terrestrial experiments. This international research collaboration employs the Plasma Krystall-4 (PK-4) on the International Space Station (ISS) to conduct experiments in microgravity which investigate the underlying physics behind self-ordering of interacting complex plasma dust particles into field-aligned chains. Scientific Goals / Objectives of the Research -- The goal of the proposed research project is to investigate the underlying physics behind self-ordering of interacting complex plasma dust particles into field-aligned chains in a microgravity environment. This goal will be accomplished by pursuing the following specific objectives: 1. Establishing the complex plasma parameters required for the formation and stability of extended field-aligned dust particle chains, 2. Measuring the grain-grain potential along and perpendicular to the ion flow direction, and 3. Studying interchain interactions and their role in the structural properties of an ordered complex plasma system. Significance of the Research -- This research will be used to develop comprehensive theories on interparticle potential and grain-plasma interactions which determine the microscopic and macroscopic properties of complex plasma, including external forces acting on the particles, the fundamental interactions between the particles, thermodynamics, and self-organization of the system. The results will directly impact research in astrophysics and planetary science (dust in protoplanetary disks; charged dust on the surface of airless bodies, such as the moon and asteroids; planetary ring systems), atmospheric physics (noctilucent clouds), fusion research (dust contamination, operating stability), materials physics (soft matter, metastable states, complex fluids, granular matter), and advanced manufacturing (semiconductor, etching, and deposition systems).
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Scientific Goals / Objectives of the Research -- The goal of the proposed research project is to investigate the underlying physics behind self-ordering of interacting complex plasma dust particles into field-aligned chains in a microgravity environment. This goal will be accomplished by pursuing the following specific objectives: 1. Establishing the complex plasma parameters required for the formation and stability of extended field-aligned dust particle chains, 2. Measuring the grain-grain potential along and perpendicular to the ion flow direction, and 3. Studying interchain interactions and their role in the structural properties of an ordered complex plasma system. Significance of the Research -- This research will be used to develop comprehensive theories on interparticle potential and grain-plasma interactions that determine the microscopic and macroscopic properties of complex plasma, including external forces acting on the particles, the fundamental interactions between the particles, thermodynamics, and self-organization of the system. The results will directly impact research in astrophysics and planetary science (dust in protoplanetary disks; charged dust on the surface of airless bodies, such as Mars, the moon and asteroids; planetary ring systems), atmospheric physics (noctilucent clouds), fusion research (dust contamination, operating stability), materials physics (soft matter, metastable states, complex fluids, granular matter), and advanced manufacturing (semiconductor, etching, and deposition systems). It also directly impacts lunar dust research including the current problem of dust mitigation on the surface of the moon, which threatens future remote or human exploration. Impact on the Development of Human Resources Over the past year, two undergraduate students and five graduate students were directly involved in the experimental portion and five undergraduate and three graduate students were directly involved in the numerical portion of the PK-4 project at Baylor. Two of the five undergraduate students mentioned are members of under-represented minorities in physics and three of the graduate students are female and now using the code in their graduate research. As part of this project, each of these students gained experience in either experimental or numerical skills as well as practice in analyzing data, writing, and presenting papers. Specifically, the experimental students learned required safety skills (i.e., laser safety, basic operating skills in HV, lasers, vacuum systems, and diagnostics) while the numerical students gained experience in testing and developing code, learned how to program using CUDA, and sharpened their skills in C++ and Matlab. (Each of these programming languages is listed among the top twenty programming languages for 2017 by the Institute of Electrical and Electronics Engineers (IEEE).) Impact on Professional Development – All of the students listed above gained experience in both technical writing and speaking with their research presented at the 61st Annual Meetings of the APS-DPP (American Physical Society-Division of Plasma Physics), the 2019 GEC (Gaseous Electronics Conference) meeting, the 2019 International Conference on Plasma Science (ICOPS), the 2019 NASA Fundamental Physics and Quantum Technology Workshop, and the 2019 Freudenberg Sealing Technologies (FST) meetings in Germany. Additionally, students presented their results as part of the Baylor Physics Department colloquium series through poster presentations during Undergraduate Research and Scholarly Achievement (URSA) Scholars Week. Additionally, over the past year CASPER students (both graduate and undergraduate) have had the opportunity to attend meetings with the Principal investigator (PI) and Co-Is from all over the world (Peter Hartmann, Hubertus Thomas, Vladimir Nosenko, Marlene Rosenberg) either at Baylor or at various conferences attended by the PI and students. Additional meetings with Ed Thomas (Auburn) and Jeremiah Williams (Wittenberg) and their students have also been

held both at conferences and at Baylor.

Task Progress:

Research Accomplishments
Experimental (Baylor) - The PK-4 BU (developed as part of this project) is currently the only device in the world which allows researchers to identify new parameter regimes or experimental plans which cannot be tested on existing PK-4 ground units. The ground unit analog PK-4 BU is now in service at Baylor University with testing, initial calibration and christening complete. Calibration data has now also been compared successfully to initial calibration tests run on the PK-4 ISS.

Specifically, over the past year (2019), the PK-4 BU was calibrated against data collected by both the PK-4 ISS and the PK-4 ground unit at DLR (German Aerospace Center) Munich. Additionally, the PK-4 BU was calibrated against initial ground test data collected during the construction of the PK-4 ISS (before launch) to verify that both PK-4 BU Langmuir probe data and LIF (logical interface) data agreed with the final calibration data produced before the PK-4 ISS was installed on board the ISS. In both cases this data was collected with the help of our collaborators, with the LIF data collected in collaboration with Caltech and the Langmuir probe data collected in calibration with the Institute of Space Systems (IRS) at the University of Stuttgart.

Experimental runs against recent PK-4 ISS data have now been completed to identify appropriate operating parameter spaces specific to this project. This data allowed the PK-4 BU to be employed by researchers from Baylor, Auburn, and DLR to coordinate two experimental campaigns which have now been completed using the PK-4 ISS. These campaigns (Campaign 7 and Campaign 9) were accepted for flight and flown on the PK-4 ISS in November of 2019 and February of 2020. The coming year (2020) will focus on non-linear wave (soliton) formation and propagation in 2D dusty plasma and dust particle lane formation in DC discharges related to the PK-4 ISS.

Experimental (DLR) – Collaboration with DLR personnel continues to occur on a regular basis with recent discussions centering on vetting and analysis of PK-4 ISS data collected during Campaign 7 (2019) as well as planning and final scripting of Campaign 9 (scheduled for flight in February of 2020). The data for Campaign 7 has now been delivered to the PIs at Auburn and Baylor (in December 2019) and is being processed so that analysis can begin. Discussion of this data at greater length will occur at the next FST meeting in Munich.

Additionally, an extensive search of the latest ISS PK-4 experimental data has been conducted (with the help of DLR personnel and CASPER personnel) to look for conditions conducive to the formation of particle strings. The most recent set of experiments on this subject was flown by DLR and that data is now being examined within CASPER.

To date, the ongoing analysis described above by CASPER (Baylor and DLR) personnel has identified (1) various regimes within the PK-4 where particle strings can form, and (2) the coexistence of strings comprised of particles of varying sizes. Both of these have been verified on the PK-4 BU device and additional analysis is underway.

Numerical (Baylor) - An entirely new code has been written and tested to model the ion-dust interactions within the PK-4 BU analog device. The code is written in C++ with CUDA to make use of the parallel processing power of Graphical Processing Units (GPUs). The code makes use of boundary conditions for the plasma provided by a PIC-MCC code used to model the PK-4 DC discharge.

The IonWake simulation is a MD simulation of ion and dust dynamics, each resolved on their own timescale. The charging of the dust is calculated self-consistently within the ion wake. The dynamics of the ion includes calculations of ion-ion accelerations, ion-dust accelerations, ion-neutral collisions, ions insertions (assuming a shifted Maxwellian velocity distribution incident on the simulation boundary), and ion deletions due to leaving the simulation boundary or impacting the dust particle. The dust dynamics includes calculations of charge, ion-dust accelerations, dust-dust accelerations, external electric fields present in the DC plasma, additional confining electric fields, gas drag, and a thermal gas bath.

The code has been tested for configurations of dust strings under a variety of gas pressures and ion flow velocities, set by the magnitude of the current applied to the polarity switching in the PK-4 experiment. Numerical (Institute for Solid State Physics and Optics, Wigner Research Centre for Physics) - A 2D, cylindrical, particle in cell code including Monte Carlo collisions (PIC/MCC) is being used in conjunction with Baylor numerical BOX_Tree models to examine gas discharge simulations for parameters matching experimentally identified PK-4 ISS and PK-4 BU operating regime data. The current version of the code is capable of accurately describing the gas-phase processes and relevant surface processes on the level of elementary electron-atom and ion-atom collisions. The simulation includes the collisional cross section set published by Biagi (v7.1), which includes electron – neon elastic scattering, ionization and excitation to 9 different (including 2 metastable) energy levels. Ion – atom collisions include isotropic elastic and symmetric charge transfer collisions. Metastable density distribution calculation and Penning ionization are included in the model. Surface processes include electron reflection and ion induced secondary electron emission from the electrodes and surface charging on the glass wall. The results of the simulations provide quantitative information of the charged particle densities, fluxes, electric field and energy distributions as well as distribution of specific collisional events, like the ionization and excitation source functions. Implementation of this model on GPU accelerated architecture turned out to be of key importance to achieve feasible simulation times.

The coming year (2020) will focus (in addition to the above) on non-linear wave (soliton) formation and propagation in 2D dusty plasma and dust particle lane formation in DC discharges related to the PK-4 ISS.

Numerical (UCSD-University of California, San Diego) – Sound speed in Yukawa systems across coupling regimes.

The UCSD PI continues to collaborate on a theoretical study of the behavior of the sound speed in a Yukawa one component plasma (YOCP), which is often used as a model for a dusty plasma. A YOCP is a system of charged particles interacting via a pairwise Yukawa (screened Coulomb) potential, which is characterized by the screening parameter, κ , and the Coulomb coupling strength, G. This study focused on the long wavelength behavior of the longitudinal acoustic mode, where the linear term in the dispersion (i.e., $\omega = sk$) defines the sound speed s. The behavior of the sound speed from the weak through the strong coupling regimes was examined by analyzing the dynamic structure factor. Five domains in the (κ , G) parameter space were delineated in which the physical behavior of the YOCP exhibits different features. Damping of the mode was also considered. Theoretical results based on various models were compared with MD computer simulation results in order to determine which model provides the best agreement with MD data in the different domains. The study has relevance to dust acoustic wave experiments under

microgravity conditions, where the behavior of the waves in the strongly coupled liquid phase may be studied.

Nonlinear development of ion-dust streaming instability: Dust acoustic (or dust density) waves have been observed in experiments both in the PK-4 BU and the PK-4 ISS. It is thought that self-excited dust acoustic waves can be driven by the flow of ions relative to the dust. The resulting dust density waves are often observed to be of high amplitude. Waveforms have been reported to have nonlinear structures such as cnodial wave forms, or waveforms with sharper crests and flatter troughs as compared with sinusoidal. The UCSD PI, together with Dr. K. Quest at UCSD and an undergraduate student Amir Levine, are investigating possible conditions under which these types of waveforms may arise in the nonlinear development of the ion-dust streaming instability. One condition appears to be that the plasma is highly collisional, where the ion and dust collision frequencies are a significant fraction of their corresponding plasma frequencies. This may be relevant to microgravity experiments or laboratory experiments with high gas pressure. One dimensional PIC simulations are ongoing, which take into account collisions of ions and dust with neutrals, and a background electric field that drives the ion flow, which is on the order of a few times thermal. Preparation of a paper, “Simulations of the ion-dust streaming instability in a highly collisional plasma,” has begun.

Waves in 1D ER dusty plasma in liquid-like phase: The UCSD PI continues to investigate the dispersion relation of waves in a somewhat disordered, liquid-like phase in a one-dimensional (1D) electrorheological (ER) dusty plasma. The model is a 1D chain of dust in the x-direction, confined by a parabolic potential in the y-direction. The binary interaction between dust particles is a combination of a Yukawa potential and an effective dipole interaction. The dipole interaction arises from ion wakes on opposite sides of the dust particle due to an AC electric field that induces subthermal ion flow (Ivlev et al., 2008, Phys. Rev. Lett. 100, 095003). This model may be relevant to microgravity experiments that use AC electric fields to keep the dust in the camera field-of- view. Using the quasi-localized charge approximation (see e.g., Donko et al, 2008, J. Phys.: Condens. Matter 20, 413101) and a pair correlation function provided by Prof. Matthews at Baylor, both longitudinal and transverse mode dispersion relations were found for a set of parameters. This could be done for other sets of parameters where the ion flow is subthermal. An outstanding theory question remains however -- what interaction potential to use when the ion flow speed is thermal to slightly superthermal, which may be relevant to certain sets of microgravity plasma parameters.

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